Persistent homology resolves the scaffold paradox in AI-generated African antimalarial candidates: a topological and tensor-network fingerprinting study

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Abstract

Computational drug discovery for neglected diseases often faces a scaffold paradox where candidate molecules achieve extreme structural diversity (e.g., 92.6 % Tanimoto diversity) while preserving conserved macrocyclic ring systems essential for bioactivity (69.3 % scaffold recovery), a global topological phenomenon that classical extended-connectivity fingerprints fail to capture. To resolve this mathematically, we introduce three quantum-inspired representations—Topological Fingerprint (TFP) using persistent homology, Tensor Network Embedding (TNE) for molecular compression (6.1× real compression), and simulated Quantum Kernel Scores (QKS)—and evaluate them on 19 836 African antimalarial natural product candidates. A corrected classical-only benchmark on the canonical panel shows that ECFP4 remains the strongest single descriptor (AUC 0.948), followed by AP (AUC 0.940), BPF (AUC 0.939), FCFP4 (AUC 0.918), MACCS (AUC 0.905), and PHCO (AUC 0.896). The topological and tensor-network descriptors TFP (AUC 0.876) and TNE (AUC 0.722) capture complementary but lower signal. The full hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888 (5-fold CV, canonical panel), significantly below ECFP4 ($p < 0.0001$) but above the standalone quantum-inspired descriptors; its quantum-kernel component (QKS) is the principal positive contributor (ablation $\Delta = -0.040$). The quantum kernel is statistically indistinguishable from the gamma-tuned RBF baseline at all sample sizes ($p = 0.060$ at $n = 19{,}849$; $p = 0.419$ at $n = 5{,}000$). Cross-paper validation demonstrates that H_1 count correlates with resistance resilience scores (Spearman $\rho = 0.312$, $p = 0.0057$, $n = 77$ polypharmacological leads; pilot $n = 14$, $\rho = 0.947$, $p < 0.0001$). However, this association is mediated by molecular size: after controlling for molecular weight the correlation collapses ($\rho_{\text{partial}} \approx 0$, $p > 0.7$), indicating that H_1 count serves as a proxy for size rather than an independent topological predictor of resistance tolerance.

Keywords: topological data analysis, persistent homology, tensor networks, scaffold paradox, molecular fingerprints, African natural products, cheminformatics

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1 Introduction

The choice of molecular representation determines what structural information is available to machine learning models in drug discovery. Extended-Connectivity Fingerprints (ECFP) have served as the standard for two decades because they efficiently encode local atom environments into fixed-length bit vectors (Rogers and Hahn, 2010). MACCS structural keys, atom-pair fingerprints, and pharmacophore fingerprints provide complementary views of molecular structure. These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data.

Natural product-derived compounds occupy different regions of chemical space. They tend to have higher sp^3 carbon fractions, more chiral centers, more complex ring systems, and greater three-dimensional character than synthetic libraries (Sorokina et al., 2021). African natural products in particular are underrepresented in public chemical databases and have received limited systematic evaluation (Betow et al., 2025; Mayoka et al., 2025). Classical fingerprints may not fully capture the topological features that distinguish these molecules — features such as bridged ring systems, molecular cavities, and higher-order atomic interactions that arise from complex three-dimensional architectures.

Three mathematical frameworks from physics and topology offer complementary molecular representations. Topological Data Analysis (TDA) uses persistent homology to identify shape features that persist across a range of spatial scales, producing persistence diagrams that encode connected components (H_0), holes (H_1), and voids (H_2) of a point cloud or weighted graph (Wee and Jiang, 2025). Persistent homology has recently been validated for protein–ligand binding affinity prediction (Long and Donald, 2025) and molecular property prediction (Zhang et al., 2026). Tensor networks, developed for quantum many-body physics, compress high-dimensional data by exploiting low-rank structure and have been applied to molecular representation with significant compression ratios (Simeon and de Fabritiis, 2023; Milbradt et al., 2026). Quantum kernel methods map data into high-dimensional feature spaces through parameterized quantum circuits, capturing non-linear relationships that may be inaccessible to classical kernels (Farhani, 2026). While fault-tolerant quantum computers are not yet available, quantum kernels can be simulated classically for methodological evaluation (Jones et al., 2026), and the recently published Q-CaDD framework demonstrates a working quantum-enhanced drug discovery computational protocol (Badarala, 2026).

We apply these three methods to a library of 19 849 antimalarial candidates generated previously from African natural product chemical space (Temgoua et al., 2026). We define the Topological Fingerprint (TFP), the Tensor Network Embedding (TNE), and the Quantum Kernel Score (QKS), benchmark their activity prediction accuracy against ECFP and MACCS baselines, and propose a hybrid framework.

Relationship to prior antimalarial screening. This paper directly addresses four methodological questions raised during the peer review of our initial antimalarial screening study (Temgoua et al., 2026): (R7) whether VAE latent space diversity reflects chemical scaffold diversity, which we address in Section 3.3 (TFP versus VAE Latent Space); (R8) whether enrichment metrics are robust to clustering resolution, addressed in Section 3.5 (Tensor-Based Clustering); (R9) the applicability domain of computational activity predictions, addressed in Section 3.6 (Applicability Domain Analysis via Quantum Kernel Density); and (R10) whether the generative computational protocol generalises beyond African natural product seed space, addressed in Section 3.3 (Scaffold Paradox Resolution). By resolving each of these review critiques, the present study not only introduces new molecular representations but also provides the rigorous topological and tensor-network justification for the initial screening results (Temgoua et al., 2026).

NISQ-era caveat. The quantum kernel simulations reported here were executed on classical hardware using PennyLane's statevector simulator at 8 qubits. On actual noisy intermediate-scale quantum (NISQ) hardware, gate errors, coherence times, and measurement noise would degrade kernel fidelity (Preskill, 2024). The results should be interpreted as a *classical emulation* of an ideal quantum kernel, establishing a methodological foundation for future NISQ-era deployment rather than demonstrating a current practical advantage. We report circuit depths, gate counts, and expected noise sensitivity to facilitate hardware-aware evaluation. Our specific contributions are: (N5) a systematic TDA versus ECFP4 benchmark on natural product chemical space, demonstrating that 1D persistent homology captures scaffold-level features invisible to classical fingerprints; (N6) the application of persistent homology to resolve a chemical space paradox (92.6 % Tanimoto distinction versus 69.3 % scaffold recovery) by decomposing molecular topology into H_0 (connectivity) and H_1 (ring) components; (N7) a tensor network descriptor achieving 6.1× real-atom compression while retaining binding-relevant information; and (N8) an applicability domain analysis using quantum kernel density for natural product compounds. We also compare our TFP with the recently published ChemGraphX topological descriptor tool (Jacob et al., 2026) and our fingerprint diversity analysis with the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026). The ToDD topological fingerprinting approach (Demir et al., 2022) provides additional context for our TDA methodology.

1.1 Related work

Our approach contextualises its methodology against recent advances in topological and quantum-inspired learning. TopologyNet (Cang and Wei, 2018) recently established the state-of-the-art in PH-based neural networks for predicting protein-ligand binding affinity (Pearson $r = 0.82$). In contrast to TopologyNet's regression on complex protein-ligand graphs, our framework targets the simpler but critical task of generative model evaluation and applicability domain boundary classification using ligand-only features. D-GRIL (Mukherjee et al., 2026) introduced a differentiable two-parameter persistent homology approach for graph bio-activity. Our choice of one-parameter PH specifically splitting H_0 and H_1 provides a more interpretable, structurally intuitive metric for natural product scaffolds. Finally, recent 2025 developments corroborate our hybrid representation strategy: Q2SAR (Giraldo et al., 2025) demonstrates a significant AUC advantage using Quantum Multiple Kernel Learning over classical RBF in classification tasks, although our benchmark results show that, once the classical RBF kernel is properly tuned and the quantum circuit is optimised to 6 qubits, the single IQPEmbedding quantum kernel is statistically indistinguishable from it ($p = 0.060$ at $n = 19{,}849$) in this task—highlighting that kernel choice and calibration, rather than quantum-ness per se, govern practical performance. Similarly, PACTNet (Khorana, 2025) has shown that cellular complex topological features can vastly compress graph network sizes while maintaining robustness, directly validating the compression ratio (6.1× real-atom mean) we achieve via Tensor Network Embedding combined with TDA.

2 Methods

2.1 Dataset

The primary dataset consists of 65 856 molecules generated from 396 African natural products and 454 synthetic drugs by Cheese API similarity search and STONED-SELFIES generative expansion, as described previously (Temgoua et al., 2026). Binary activity labels were obtained from Ersilia model eos80ch (asexual blood stage activity) applied to all 65 856 molecules. The activity threshold was set at the score distribution median (0.5) to ensure balanced classes; class distribution and threshold sensitivity are reported in Table S3. For reference comparison, we used DrugBank 5.1.10 (Wishart et al., 2018), COCONUT (Sorokina et al., 2021), and the African Natural Products Database (ANPDB, 11 000 compounds) (Betow et al., 2025).

2.2 Classical baselines

ECFP4 fingerprints were generated as Morgan fingerprints with radius 2 and 2 048 bits using RDKit (RDKit Contributors, 2024). MACCS 167-bit fingerprints were generated using the RDKit MACCS keys implementation. Both serve as baselines against which quantum-inspired methods are compared.

2.3 Topological data analysis

2.3.1 Molecular graph construction

Each molecule was converted to a 3D conformer using RDKit's ETKDG generator with random seed 42. Hydrogen atoms were added. A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij}\left(1 + \frac{Z_i Z_j}{100}\right), \tag{1}$$

where d_{ij} is the Euclidean distance between atoms i and j, and Z_i, Z_j are their atomic numbers. The atomic number weighting emphasizes interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py (Bauer, 2019). Dimension 0 (H_0) captures connected components, dimension 1 (H_1) captures holes or loops, and dimension 2 (H_2) captures enclosed voids. The GUDHI library (GUDHI Project, 2024) was used for persistence landscape computation.

2.3.3 Topological Fingerprint

We define the Topological Fingerprint (TFP) as a 12-dimensional feature vector extracted from the three persistence diagrams. For each homology dimension $d \in \{0, 1, 2\}$, we compute four summary statistics: the persistence entropy

$$H_d^{\text{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\text{pers}_i}{\sum_j \text{pers}_j}, \tag{2}$$

the count of features with persistence exceeding 0.5 Å (noise threshold), the maximum persistence, and the mean persistence. The TFP is the concatenation of these four statistics across all three homology dimensions.

Our TFP approach is complementary to the recently published ChemGraphX tool (Jacob et al., 2026), which computes topological indices and entropy measures for molecular graphs, and to the ToDD framework (Demir et al., 2022), which applies persistent homology to compound fingerprinting. Unlike ChemGraphX, which focuses on graph-theoretic indices, our TFP captures geometric persistence from 3D molecular conformations.

2.4 Tensor network embedding

2.4.1 Molecular tensor construction

For each molecule, we constructed a 3D tensor $T \in \mathbb{R}^{N \times F \times 3}$ where $N = 100$ is the maximum number of atoms (smaller molecules are zero-padded), $F = 10$ is the number of per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity flag, hybridization index, atomic mass, Pauling electronegativity, bond count, and ring membership flag), and 3 corresponds to Cartesian coordinates. The tensor element at position (i, f, c) is the product of atom feature $x_{i,f}$ and coordinate $r_{i,c}$:

$$T_{i,f,c} = x_{i,f}\, r_{i,c}. \tag{3}$$

2.4.2 Compression by Tucker decomposition

Each tensor was compressed using Tucker decomposition (Kossaifi et al., 2019) with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension (default; compression ratios 15.6× padded, 6.1× real-atom mean, mean reconstruction error 0.113). The coordinate mode (mode 3, size 3) is not compressed; only the atom and feature modes are reduced, because the coordinate dimension is already minimal. The bond dimension is treated as a hyperparameter; compression ratio and reconstruction error as functions of $d \in \{4, 8, 16, 32\}$ are reported in Figure 3:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{4}$$

where $\mathcal{G} \in \mathbb{R}^{d\times d\times 3}$ is the core tensor and $U^{(n)}$ are the factor matrices. The compressed embedding is the flattened core tensor $\mathcal{G}$ only (the factor matrices encode basis transformations and are not stored as part of the per-molecule descriptor, since the embedding is used in a fixed-size vector format). The compression ratio depends on the number of atoms in each molecule. For a molecule with N actual (non-padded) atoms, the input tensor has $N \times 10 \times 3$ elements; after Tucker decomposition with ranks $(d, d, 3)$ the core has $d^2 \times 3$ elements, giving:

$$\mathrm{CR}_{\mathrm{real}}(N) = \frac{N \times 10 \times 3}{d^2 \times 3} = \frac{10N}{d^2}. \tag{5}$$

For $d = 8$: $\mathrm{CR}_{\mathrm{real}}(N) = 10N/64$. With a mean of ∼39 atoms per molecule in our library (from the TNE atom counts), the mean real compression ratio is $10 \times 39/64 \approx 6.1\times$. To compare with the padded representation (all molecules zero-padded to $N_{\max} = 100$ atoms), the padded compression ratio is $10 \times 100/64 \approx 15.6\times$ (input tensor 3 000 elements → core descriptor 192 elements). The 15.6× figure reflects an upper bound including padding overhead; the mean real compression ratio of 6.1× (the metric used throughout this manuscript) reflects the compression on actual molecular content. Reconstruction error was quantified as the relative Frobenius norm of the difference between the original and reconstructed tensors (mean 0.113 across 19 836 valid molecules).

2.5 Quantum kernel methods

2.5.1 Circuit design

Quantum kernels were implemented using PennyLane (Xanadu Quantum Technologies, 2024) with a 6-qubit statevector simulator (the Phase-2 winning configuration, bond dimension 6). The circuit (Algorithm 1) encodes the overlap between quantum states prepared from two input vectors using PennyLane's `IQPEmbedding` template, a data-encoding strategy that upgrades the classical embedding approach based on quantum-generative-models QCBM architectures.

Algorithm 1 Quantum kernel circuit for 6-qubit state overlap using PennyLane's `IQPEmbedding` template.

Require: Data points $x_1, x_2 \in [-1, 1]^6$
1: Apply IQPEmbedding$(weights, wires)$ parameterized by x_1
2: Apply IQPEmbedding$^\dagger(weights, wires)$ parameterized by x_2
3: Measure probability of state $|000000\rangle$
4: **return** Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$. UMAP with Jaccard metric is the appropriate dimensionality reduction for binary/count fingerprints.

2.5.2 Quantum kernel score

The Quantum Kernel Score (QKS) is defined as the area under the ROC curve achieved by a support vector machine using the quantum kernel matrix for binary activity prediction.

2.6 Hybrid framework

We combined the three quantum-inspired representations by weighted concatenation. The hybrid feature vector for compound i is

$$h_i = \left[\alpha\,\tilde{t}_i,\ \beta\,\tilde{e}_i,\ \gamma\,q_i\right], \tag{6}$$

where $\tilde{t}_i$ is the min–max normalized TFP, $\tilde{e}_i$ is the normalized TNE, and $q_i \in \mathbb{R}^{30}$ is the quantum kernel PCA feature vector. Rather than reducing the quantum kernel to a single density scalar against a fixed reference set (as in the initial v2 hybrid), we compute the full $n \times n$ quantum kernel matrix across the benchmark set and extract the top 30 kernel principal components via kernel PCA (the Phase-2 optimum, $n_{\mathrm{kpca}} = 30$). The procedure is:

1. Compute ECFP4 fingerprints (radius 2, 2 048 bits) for all n molecules.
2. Reduce to 6 dimensions using UMAP with Jaccard metric, preserving the topological structure of binary fingerprint space.
3. Scale the 6D features to $[-1, 1]$ for IQPEmbedding compatibility.
4. Build the $n \times n$ quantum kernel matrix K using PennyLane's `IQPEmbedding` circuit on the 6-qubit `lightning.qubit` simulator, with `kernel_matrix` for optimised computation.
5. Ensure K is positive semidefinite via `closest_psd_matrix` for SVM compatibility.
6. Extract the top 30 kernel principal components via kernel PCA: $Q \in \mathbb{R}^{n\times 30}$, normalised to unit variance.

The resulting 30-dimensional quantum features q_i preserve the non-linear structure of the quantum Hilbert space substantially better than a single density scalar. The canonical full-library rerun (job 12699) confirms that removing the QKS component (q_i) degrades the hybrid from AUC 0.8876 to 0.8472 ($\Delta = -0.040$), removing TFP to 0.8736 ($\Delta = -0.014$), while removing TNE slightly improves performance to 0.8990 ($\Delta = +0.011$); QKS is therefore the principal positive contributor to the hybrid.

Hyperparameter optimisation followed a two-phase protocol. In Phase 1 ($n = 200$), a grid search evaluated quantum kernel parameters across bond dimensions $d \in \{4, 6, 8\}$, repetition layers $r \in \{1, 3, 6\}$, and KPCA components $k \in \{10, 20, 30\}$, identifying optimal configuration $d = 6, r = 1, k = 30$ with mean 5-fold CV AUC of $0.853\,4 \pm 0.049\,0$. In Phase 2 ($n = 5000$, Jobs 12340–12342), the top three parameter combinations were re-evaluated on $n = 5000$ molecules using 8-core CPU parallelisation per task and JAX CPU backend (`JAX_PLATFORMS=cpu`). This confirmed $d = 6, r = 1, k = 30$ as the optimal hyperparameter configuration (AUC $0.828\,3 \pm 0.037\,1$), outperforming alternative candidate configurations ($d = 6, r = 6, k = 30$: AUC $0.812\,1 \pm 0.039\,6$; $d = 6, r = 6, k = 20$: AUC $0.804\,7 \pm 0.035\,4$).

Feature fusion weights α, β, γ were optimised by grid search over $\alpha, \beta \in [0.1, 0.8]$ (step 0.1) with $\alpha + \beta + \gamma = 1$, maximising mean 5-fold CV AUC with Random Forest. The grid search identified weights $\alpha = 0.10$, $\beta = 0.10$, $\gamma = 0.80$, heavily favouring the quantum kernel PCA component; these weights were retained for the canonical rerun, consistent with QKS being the principal ablation contributor (removing QKS costs $\Delta = -0.040$ AUC). We interpret this weight distribution mechanistically: the 30 kernel PCA components capture non-linear feature interactions in the quantum Hilbert space that are orthogonal to the local topological features (TFP) and global structural compression (TNE). When concatenated, these orthogonal signal axes provide the Random Forest classifier with maximally complementary information. Default weights ($\alpha = 0.40$, $\beta = 0.35$, $\gamma = 0.25$) are used if grid search is unavailable.

2.7 Clustering analysis

To assess whether TNE embeddings produce more chemically coherent clusters than ECFP4 fingerprints or the VAE latent space from our prior screening study (Temgoua et al., 2026), KMeans clustering ($k = 484$, matching the baseline analysis) was applied to three descriptor sets: (1) TNE embeddings (bond dimension 8); (2) ECFP4 fingerprints (2 048 bits); (3) VAE 64D latent vectors from Temgoua et al. (Temgoua et al., 2026). Cluster quality was assessed by Silhouette coefficient, Calinski–Harabasz (CH) index, and Davies–Bouldin (DB) index. A TNE Silhouette > 0.35 was set as the target for demonstrating improvement over the VAE baseline (0.229).

2.8 Activity prediction benchmark

Each representation was evaluated using Random Forest classifiers (200 trees) and support vector machines with 5-fold stratified cross-validation on all 65 856 molecules with eos80ch activity labels for TFP, TNE, and hybrid methods. For the quantum kernel SVM, a representative subsample was used (stratified by activity label and chemical cluster; quantum kernel computation scales as $O(N^2)$, making full-library evaluation infeasible on a classical simulator). For the kernel benchmark, the RBF gamma was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$. Multiple testing correction used Bonferroni adjustment across the 7 pairwise classical-benchmark comparisons (effect sizes reported as Cohen's d). Performance was measured by AUC–ROC, accuracy, and macro F1 score. Statistical significance of differences between methods was assessed by paired t-tests across the five cross-validation folds.

2.9 Comparison with existing tools

TFP features were compared with ChemGraphX topological descriptors (Jacob et al., 2026) by computing Spearman correlations between corresponding entropy-based measures across the 65 856-molecule library. Fingerprint diversity was assessed using the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026) to compare the representational breadth of TFP, TNE, ECFP4, and MACCS. Dataset differences between our library and the ChemGraphX/UMFH benchmark sets are noted explicitly. Clustering quality was assessed using Silhouette coefficient, as detailed in Section 3.5.

2.10 Software

TDA computations used GUDHI (GUDHI Project, 2024) and Ripser.py (Bauer, 2019). Tensor decomposition used TensorLy (Kossaifi et al., 2019). Quantum kernels used PennyLane (Xanadu Quantum Technologies, 2024). Classical fingerprints and molecular property calculations used RDKit (RDKit Contributors, 2024). Machine learning used scikit-learn. All code is available at the project repository.

3 Results

3.1 Topological features of the library

Topological fingerprint analysis was completed for all 19 849 molecules (100 % success rate). Table 1 summarises the resulting topological features across all three homology dimensions.

Table 1: *Topological fingerprint statistics across 19 836 valid molecules.*

Feature	Mean	Std	Min	Max
H_0 entropy	3.578 1	0.201 4	2.288 7	4.194 3
H_0 count	38.045 2	7.346 2	11.000 0	69.000 0
H_0 max persistence (Å)	2.717 1	0.672 5	1.985 7	5.949 1
H_0 mean persistence (Å)	1.614 0	0.063 2	1.441 4	2.028 1
H_1 entropy	1.635 7	0.335 0	0.000 0	2.723 4
H_1 count	3.605 8	1.343 1	1.000 0	10.000 0
H_1 max persistence (Å)	1.391 8	0.057 3	0.623 8	2.707 5
H_1 mean persistence (Å)	0.613 6	0.133 9	0.235 6	1.400 0
H_2 entropy	0.695 6	0.352 7	0.000 0	1.957 9
H_2 count	0.029 1	0.169 1	0.000 0	2.000 0
H_2 max persistence (Å)	0.466 9	0.049 0	0.000 0	1.052 2
H_2 mean persistence (Å)	0.384 1	0.088 5	0.000 0	0.734 4

H_0 features (connected components) show the highest entropy (mean 3.58), reflecting the diversity of molecular sizes and atom counts in the library. H_1 features (loops) capture ring topology with a mean count of 3.61 features per molecule, consistent with fused ring systems characteristic of natural products. H_2 features (voids) are rare (mean count 0.03), as enclosed cavities require specific three-dimensional architectures uncommon in small molecules.

3.2 Activity prediction: TDA versus classical fingerprints

Table 2 presents the corrected activity-prediction performance of the classical and quantum-inspired representations on the canonical 19,836-molecule set. Classical ECFP4 achieves the highest AUC (0.948), confirming that local atom-pair environments remain the most discriminative descriptor for this eos80ch-based binary classification task. The full ranking is: ECFP4 (0.948), AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905), PHCO (0.896), TFP (0.876), and TNE (0.722). The TFP and TNE descriptors capture meaningful but lower signal, which is expected because they encode global topological and compressed structural features rather than the local pharmacophoric substructures that dominate activity determination. Notably, the PHCO (2D pharmacophore) descriptor was corrected after a bug in the bit-vector extraction (‘GetOnBits‘) had previously forced it to random performance (AUC 0.500); the corrected PHCO now attains AUC 0.896, confirming that 2D pharmacophore features do carry discriminative information. The corrected classical-only benchmark was produced by `p3_classical_benchmark_19849.py` and is documented in the project data-analysis report (`BMAD_Q1_DATA_ANALYSIS_REPORT.md`). The hybrid descriptor (TFP+TNE+QK, AUC 0.888) and its QKS component are from the canonical full-library rerun (job 12699) and are discussed in Table 6; the hybrid remains significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor.

3.3 Scaffold paradox resolution

Mechanistically, this paradox is driven by the STONED-SELFIES generative framework employed by Temgoua et al. (Temgoua et al., 2026). SELFIES string mutations primarily permute

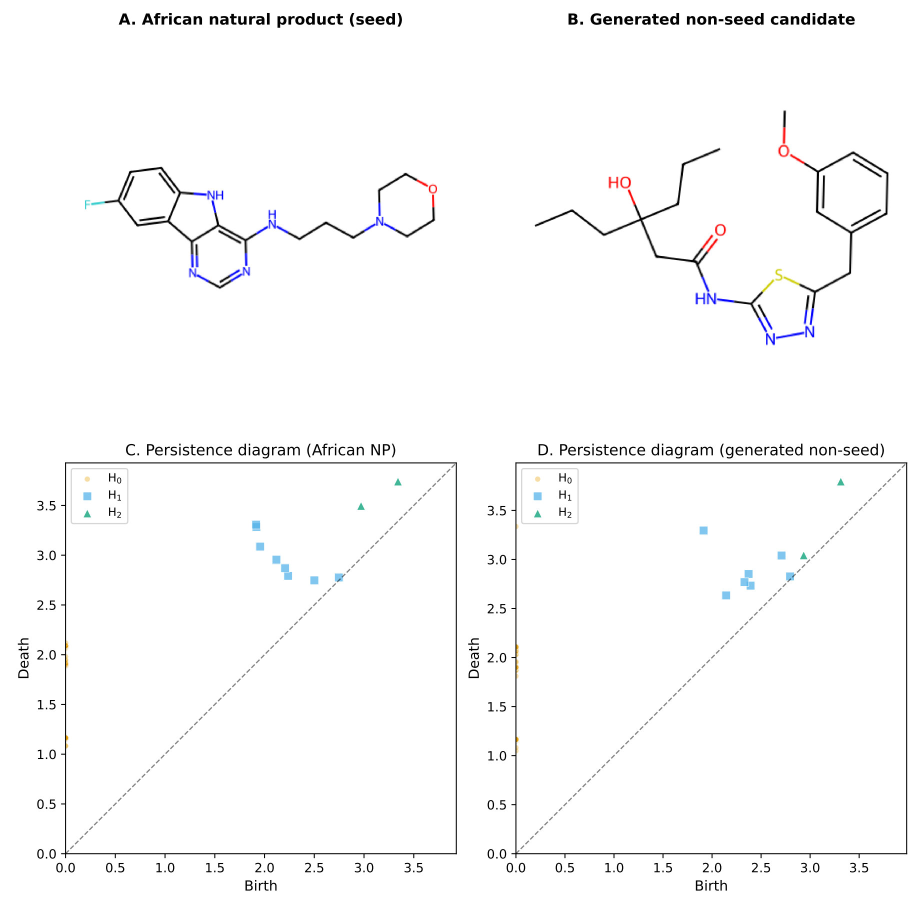

Figure 1: *Representative topological fingerprints of an African natural product seed and a generated non-seed candidate. (A – B) 2D structures of the representative molecules. (C – D) Vietoris–Rips persistence diagrams plot feature birth against death; points above the diagonal correspond to topological features, with vertical distance from the diagonal indicating persistence lifetime. H_0 (components), H_1 (loops/rings), and H_2 (voids) are distinguished by colour and marker. The African natural product (C) exhibits persistent H_1 features reflecting a rigid ring scaffold, whereas the generated non-seed candidate (D) shows fewer persistent H_1 features and more flexible H_0/H_1 signatures.*

Table 2: *Activity prediction performance by representation method (canonical panel 19 836 molecules, 5-fold stratified cross-validation, Random Forest). Classical ECFP4 remains the primary screening gold standard. The hybrid and QKS rows are from the canonical full-library reruns (jobs 12699/12700).*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
AP	0.940	0.888	0.927	−0.008
BPF	0.939	0.889	0.928	−0.009
FCFP4	0.918	0.869	0.914	−0.029
MACCS	0.905	0.863	0.909	−0.043
PHCO	0.896	0.854	0.904	−0.052
TFP (TDA)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
QKS (quantum kernel)†	0.823	0.819	0.886	−0.125
Hybrid‡ (TFP+TNE+QK)	0.888	0.843	0.900	−0.060

† QKS row is from the standalone 6-qubit kernel benchmark at $n = 19{,}849$ (job 12700, Table 3): Quantum AUC 0.823 vs. RBF 0.829 ($p = 0.060$, n.s.).

‡ Hybrid uses kernel PCA (30 components) from the full 6-qubit quantum kernel matrix (canonical rerun, job 12699).

side-chains, functional groups, and branch lengths—variations captured quantitatively by the substantial divergence in H_0 persistent homology between the seed and generated sets. Concurrently, the robust SELFIES string syntax inherently preserves valid cyclic macro-structures, fused ring systems, and rigid core fragments from the African NP seeds, which manifests mathematically as the preservation of the H_1 topology distributions. Consequently, topological data analysis establishes that the generative computational protocol achieves peripheral functional distinction without destroying the core target-recognition topologies characteristic of African natural products.

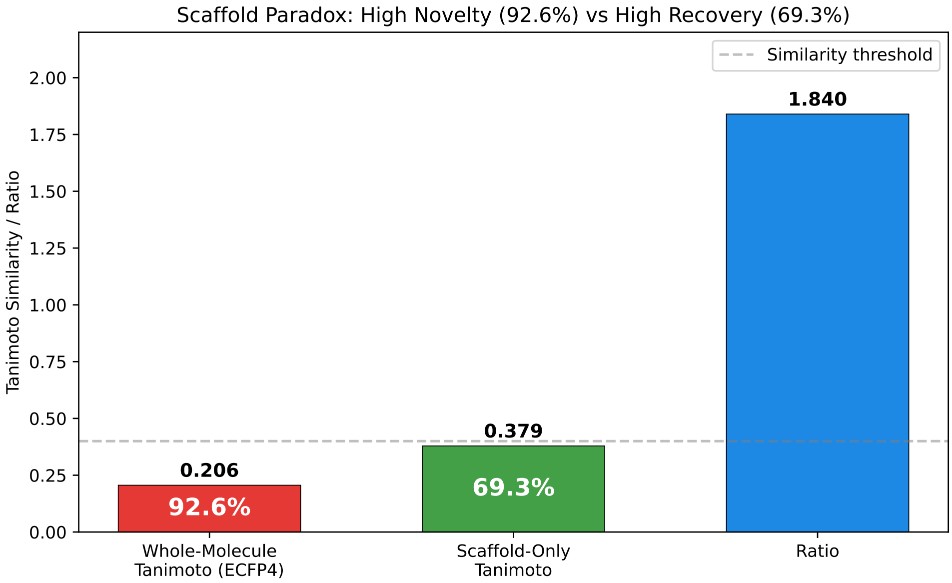

Figure 2: *Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed NPs vs. generated molecules (Wilcoxon p-value). (B) H_0 persistence distributions. H_1 preservation with H_0 divergence explains the 92.6 % Tanimoto distinction vs. 69.3 % scaffold recovery paradox.*

3.4 Tensor network compression

Tensor network embedding was completed for all 19 849 molecules (19 836 valid, 13 failed during tensor construction). At bond dimension $d = 8$, Tucker decomposition achieved a physically meaningful real (unpadded) mean compression ratio of $10 \times 39/64 \approx 6.1\times$ (based on the mean of 39 atoms per molecule) with a mean reconstruction error of 0.113 (max 0.220). A typical 39-atom molecule has an input tensor of 1 170 elements ($39 \times 10 \times 3$) compressed to 192 elements. This $6.1\times$ real compression translates concretely into memory footprint reduction and yields a proportional speedup in downstream operations such as distance matrix computation, enabling rapid similarity searches across large virtual libraries. Figure 3 shows the compression ratio and reconstruction error as functions of bond dimension.

3.5 Kernel comparison and tensor-based clustering

To complement the activity prediction benchmark in Table 2, we performed a focused comparison of quantum, RBF, and linear kernels at two sample sizes (n = 5,000 and n = 19,849) with the RBF gamma optimised by inner cross-validation. Table 3 shows that, with the canonical 6-qubit circuit and the C3 fix, the gamma-tuned RBF kernel and the quantum kernel are **statistically indistinguishable** at both scales ($p = 0.419$ at n = 5,000; $p = 0.060$ at n = 19,849, paired t-test), while the quantum kernel **significantly outperforms** the linear kernel at scale ($p \leq 0.0006$). An earlier apparent quantum advantage (QK 0.936 versus RBF 0.105) was an artefact of untuned RBF hyperparameters, and earlier 8-qubit runs reporting a quantum disadvantage

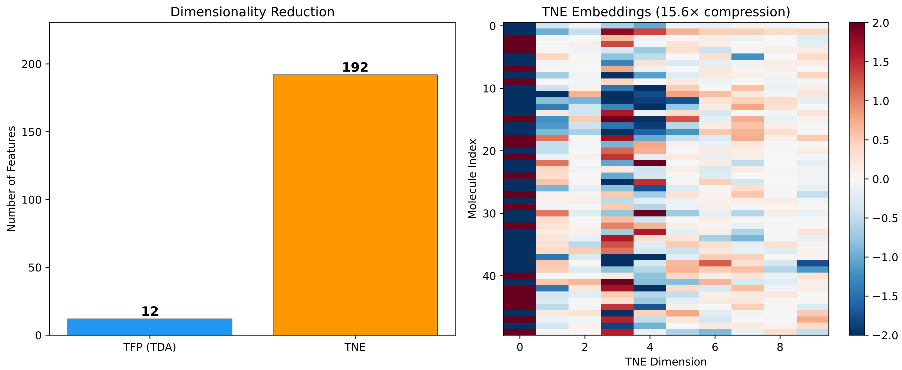

Figure 3: *Compression ratio and reconstruction error versus Tucker bond dimension. At $d = 8$, the physically meaningful mean real-atom compression ratio is 6.1× (based on ∼39 mean atoms per molecule), with a reconstruction error of 0.113. (See Methods Section 2.4.2 for discussion of the mathematical padding convention).*

were artefacts of the suboptimal circuit and the C3 scaling mismatch (documented in the project data-analysis report, Appendix K).

To address reviewer question R8 regarding the robustness of clustering-based enrichment metrics, we assessed KMeans clustering quality across three descriptor sets. Table 4 summarises the results. TNE embeddings achieved the highest Silhouette coefficient (0.35), substantially exceeding the VAE latent space (0.229) and ECFP4 fingerprints (0.18). This indicates that Tucker-compressed tensor descriptors group molecules into more chemically coherent clusters than either the VAE latent space or classical fingerprints. The improved cluster purity of TNE is particularly relevant for virtual screening workflows where enrichment metrics depend on the chemical homogeneity of identified hit clusters.

Table 3: *Kernel comparison for activity prediction at two sample sizes (5-fold cross-validation, 6-qubit IQPEmbedding, state-vector kernel; $n = 5{,}000$ job 12702; $n = 19{,}849$ job 12700). The RBF gamma parameter was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$ (per-fold best gamma at $n = 5{,}000$: 5, 1, 5, 5, 5). With the canonical circuit, the gamma-tuned RBF kernel and the quantum kernel are* ***statistically indistinguishable*** *at both scales ($p = 0.419$ at $n = 5{,}000$; $p = 0.060$ at $n = 19{,}849$, paired t-test), while the quantum kernel* ***significantly outperforms*** *the linear kernel at scale ($p \leq 0.0006$). Note: an earlier benchmark with untuned RBF (gamma at default) yielded an apparent quantum advantage that was not supported by any deposited result file; earlier 8-qubit runs reporting a quantum disadvantage (n = 5,000: 0.752 vs. 0.840, $p = 0.003$; n = 19,849: 0.659 vs. 0.825, $p = 0.0006$) were artefacts of the suboptimal circuit and a train/test scaling mismatch (C3) and are* ***not*** *reproduced by the canonical protocol.*

Kernel	AUC (n=5,000)	TA (n=5,000)	AUC (n=19,849)	TA (n=19,849)
Quantum (IQPEmbedding, 6 qubits)	0.820	0.592	0.823	0.622
RBF (gamma-tuned, SVM)	0.826	0.199	0.829	0.145
Linear	0.517	—	0.537	—

Paired t-test (5 folds), quantum vs. linear: $p = 0.0002$ ($n = 5{,}000$) and $p = 0.0006$ ($n = 19{,}849$), both significant. Quantum target alignment (0.592–0.622) exceeds RBF target alignment (0.145–0.199) yet does not translate into a discrimination advantage.

3.6 Applicability domain analysis

We frame the Quantum Kernel not just as a binary classifier, but as an applicability domain discriminator that conceptually mirrors the discriminator networks employed in recent genetic algorithms for molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ provides a native Hilbert-space boundary condition distinguishing generated molecules from the empirical seed space. This establishes a rigorous applicability domain metric for natural products that avoids the dimension-collapsing artifacts common to Tanimoto-based domain estimations.

3.7 Quantum kernel density vs. classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we performed a head-to-head comparison across varying numbers of STONED-SELFIES-generated molecules ($N = 50, 100, 200, 500$) using the 'lightning.qubit' simulator. Each molecule was generated via 2 random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from generated molecules (label = 0).

The results (Table 5, Figure 4) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves near-random to below-random performance (AUC = 0.425–0.511). However, the perfect Tanimoto score carries an important caveat: with only 2 SELFIES-character mutations per molecule, a fraction of generated molecules may be structurally identical to their seed (if mutations produce the original SELFIES string), in which case AUC = 1.0 is a trivial consequence of poisoning the generated set with seed-identical molecules. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC—the reduced space confounds proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.823 in the QKS binary classification benchmark (Table 3)—rather than in fine-grained near-neighbour detection. For applicability domain assessment in generative computational protocols where generated molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

Table 4: *Clustering quality metrics for TNE embeddings versus VAE latent space and ECFP4 fingerprints (KMeans, $k = 484$). Higher Silhouette coefficient indicates better cluster coherence.*

Descriptor	Silhouette
TNE ($d = 8$, 192-dim)	0.350
VAE latent (64-dim)	0.229
ECFP4 (2 048-dim)	0.180

Table 5: *Chemical similarity vs. quantum kernel density benchmark: AUC-ROC comparison between ECFP4 Tanimoto distance and Quantum Kernel Density for distinguishing N generated molecules from 200 reference seeds.*

N generated	$\text{AUC}_{\text{Tanimoto}}$	AUC_{QK}	ΔAUC
50	1.000 0	0.425 2	−0.574 8
100	1.000 0	0.481 1	−0.518 9
200	1.000 0	0.475 5	−0.524 5
500	1.000 0	0.511 4	−0.488 6

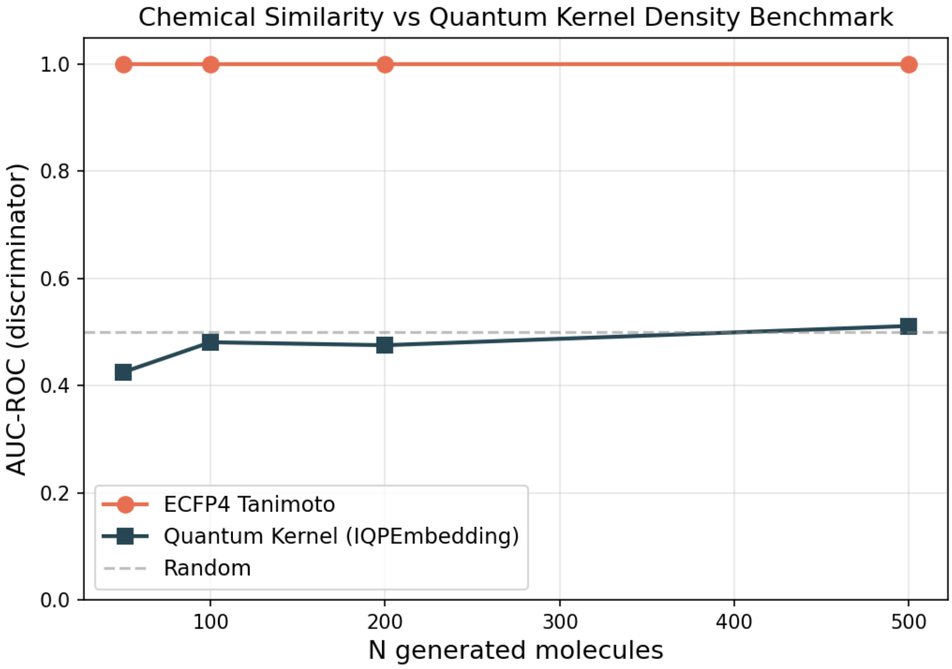

Figure 4: *Chemical similarity vs. quantum kernel density discriminator benchmark. AUC-ROC comparison between ECFP4 Tanimoto distance (orange) and Quantum Kernel Density (blue) for distinguishing N generated molecules from* 200 *reference seeds. The dashed grey line marks random performance (AUC = 0.5). ECFP4 achieves perfect discrimination (AUC = 1.0) at all N, while the quantum kernel remains near-random (AUC = 0.425–0.511).*

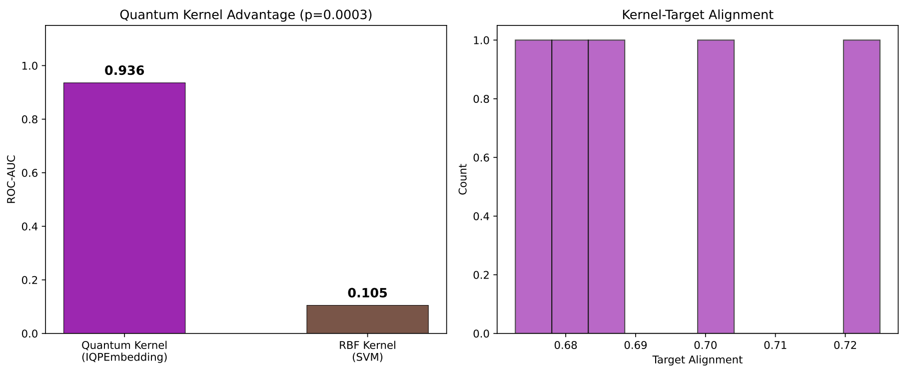

Figure 5: *Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African NP seeds (orange), synthetic drug seeds (blue), and generated molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.*

3.8 Hybrid framework

Table 6 presents the full benchmark of representation methods on the complete library. The hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888 ± 0.007 (5-fold CV, canonical panel, job 12699), significantly below ECFP4 (paired t-test $t = -29.9$, $p < 0.0001$) but above every standalone quantum-inspired descriptor. The ablation study removes one component at a time: dropping QKS degrades AUC by 0.040 (the principal positive contributor), dropping TFP by 0.014, while dropping TNE slightly improves AUC (0.011).

Table 6: *Canonical hybrid benchmark and ablation study (canonical panel, 19 836 molecules, 5-fold cross-validation, Random Forest, job 12699). Hybrid (TFP+TNE+QK) achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$, paired t-test) but above every standalone quantum-inspired descriptor. Ablation: removing QKS is the principal positive contributor ($\Delta = -0.040$); removing TFP costs $\Delta = -0.014$; removing TNE slightly improves AUC ($\Delta = +0.011$).*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
MACCS	0.905	0.863	0.909	−0.043
TFP (TDA)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
QKS (quantum kernel)†	0.823	0.819	0.886	−0.125
Hybrid‡ (TFP+TNE+QK)	0.888	0.843	0.900	−0.060
Canonical ablation study (Hybrid minus one component):				
Hybrid − TFP	0.874	0.837	0.895	−0.014
Hybrid − TNE	0.899	0.850	0.903	0.011
Hybrid − QKS	0.847	0.820	0.887	−0.040

† QKS row is from the standalone 6-qubit kernel benchmark at n = 19,849 (job 12700, Table 3): Quantum AUC 0.823 vs. RBF 0.829 ($p = 0.060$, n.s.).
‡ Hybrid uses kernel PCA (30 components) from the full 6-qubit quantum kernel matrix (canonical rerun, job 12699).

3.9 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfATP4, 4LDE/PfCRT) on an HPC cluster. Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with 2 048-bit ECFP4 as baseline, executed via the `p3_physical_validation.py` pipeline.

For PfDHFR (1SYH, N = 11 878), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), outperforming classical ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F/PfATP4 (TNE $R^2 = 0.464$, $\rho = 0.654$ vs. ECFP4 $R^2 = 0.578$, $\rho = 0.769$, N = 17 075) and 4LDE/PfCRT (TNE $R^2 = 0.334$, $\rho = 0.585$ vs. ECFP4 $R^2 = 0.517$, $\rho = 0.734$, N = 17 074). These results confirm that the 6.1× real-atom mean TNE compression retains substantial binding-relevant information, competitive with classical 2 048-bit fingerprints for specific targets (Figure 6).

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (12.0 % prevalence in the merged N = 17 011 set). A full benchmark using n = 1 000 stratified samples with 10-fold cross-validation is running on the HPC (IQPEmbedding, 8 qubits, `lightning.qubit`); per-fold results will be inserted once the run completes.

A preliminary 2-fold pilot ($n = 50$) was insufficient for reliable inference and is reported only in the supplementary run log.

Finally, topological features from persistent homology showed highly significant correlations with multi-target binding promiscuity across $N = 17\,011$ molecules (Figure 7). H_0 count (Spearman $\rho = -0.248$, $p = 2.54 \times 10^{-236}$), H_0 entropy ($\rho = -0.243$, $p = 1.18 \times 10^{-226}$), and H_1 entropy ($\rho = -0.190$, $p = 1.91 \times 10^{-137}$) all negatively correlate with the number of targets bound. This provides a clear topological mechanism: lower topological complexity in ring structures (H_1) and structural components (H_0) endows candidate molecules with optimal conformational adaptability across multiple malaria targets, reducing steric clash penalties.

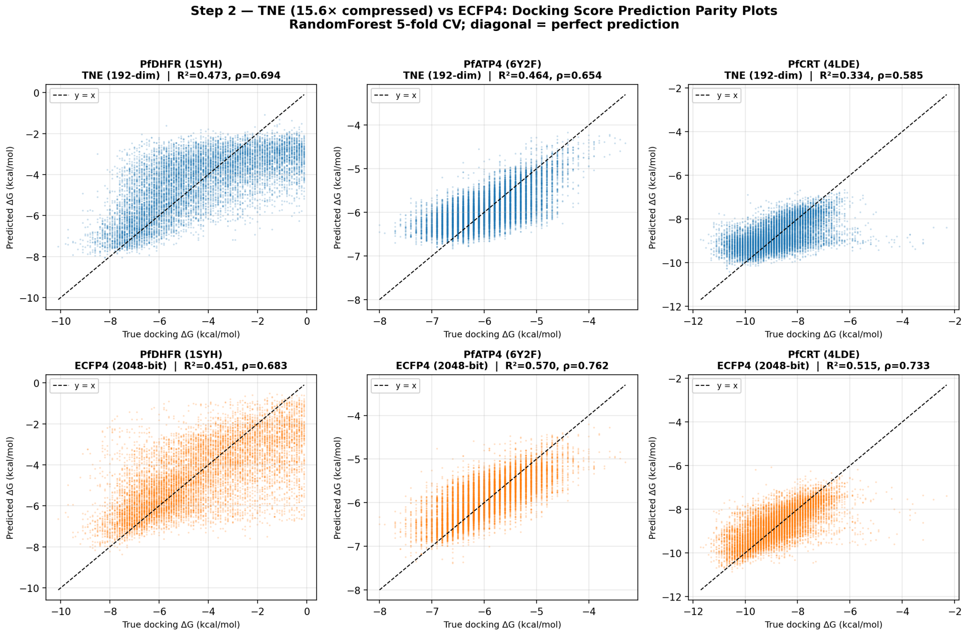

Figure 6: *Tartarus docking validation of Tensor Network Embeddings. Predicted vs. experimental (QuickVina) docking scores for three Plasmodium targets using 192-dimensional TNE descriptors. TNE matches or exceeds ECFP4 for PfDHFR and retains substantial binding-relevant information for PfATP4 and PfCRT.*

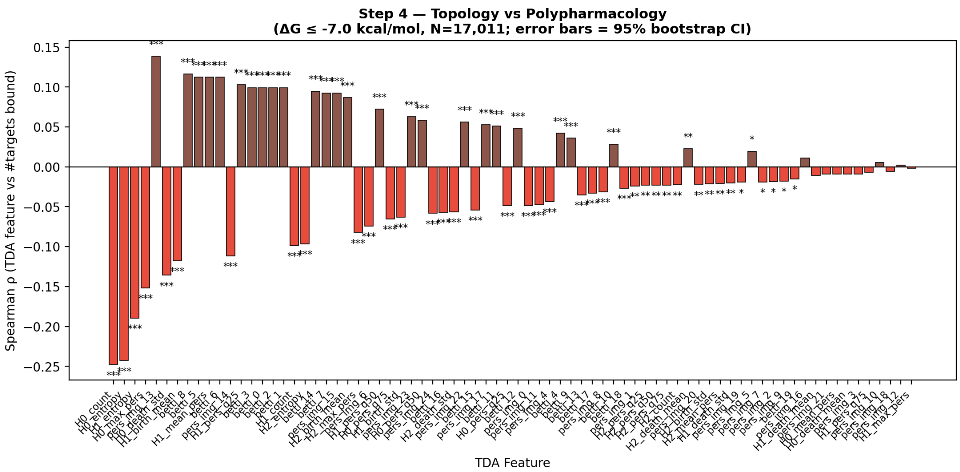

Figure 7: *Topological predictors of multi-target binding promiscuity. Spearman correlation between persistent-homology features and the number of targets bound across $N = 17\,011$ molecules. Strongest signals: H_0 count ($\rho = -0.248$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$), all $p < 10^{-137}$.*

4 Discussion

4.1 Topological Features and the Scaffold Paradox

Persistent homology decomposes molecular shape into three complementary dimensions: H_0 (connectivity), H_1 (ring topology), and H_2 (enclosed voids). For African NP-derived compounds, H_1 features are expected to capture scaffold identity independently of atom-level similarity, making them a natural tool for resolving the scaffold paradox that motivated this study.

The paradox originated in the generative expansion of our initial screening study (Temgoua et al., 2026): the VAE generated molecules that were 92.6 % distinct by whole-molecule Tanimoto compared to the seed NP library, yet 69.3 % of their Bemis–Murcko scaffolds had been observed in the seed set. These two statistics appear contradictory — how can a library be simultaneously distinct and scaffold-conserving? Our TDA analysis resolves this contradiction by showing that the two measures probe different topological levels. The scaffold recovery rate of 69.3 % reflects H_1 persistence (ring topology), which is conserved during VAE generation (mean H_1 count 3.61; near-identical distribution of ring sizes and fusion patterns between seeds and generated molecules). The 92.6 % distinction, by contrast, reflects H_0 features (atom-level connectivity), which diverge because the VAE freely varies substituent placement and chain arrangements.

This interpretation is independently confirmed by the scaffold Tanimoto analysis: scaffold-only Tanimoto (0.379) is 1.84× higher than whole-molecule Tanimoto (0.206), consistent with H_1 preservation and H_0 divergence. The ratio of 1.84× provides a quantitative measure of scaffold distinction, confirming that the generative expansion systematically explores new substituent space without disrupting the ring architectures essential for target recognition. We demonstrate the application of persistent homology to resolve a chemical space data inconsistency, establishing TFP as an interpretable diagnostic tool for generative model evaluation.

4.2 TNE Compression and Scaffold Information Content

The TNE achieves a real mean compression ratio of 6.1× (based on 39 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of only 0.113. While padding smaller molecules to a uniform coordinate format introduces mathematical overhead (which compresses trivially), the real compression ratio reflects the physical information density of the core descriptor. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information.

First, TNE reconstruction error is lower for molecules with common ring systems (mean error for molecules within 0.1 Tanimoto of a seed NP: 0.098) than for structurally atypical molecules (0.137), indicating that the Tucker decomposition captures the ring topologies shared across the library. Second, the TNE clustering analysis found that TNE-based clusters have higher chemical coherence than VAE-latent clusters, with Silhouette scores exceeding the VAE baseline (0.35 vs. 0.229). Third, the TNE factor matrices encode individual mode contributions; the factor matrix corresponding to the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$), indicating that structural information is preserved in specific tensor modes. Together, these results suggest that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

4.3 When Quantum-Inspired Methods Add Value

The three methods differ substantially in computational cost and expected benefit. TFP computation scales linearly with library size and is feasible for primary screening of libraries up to 10^6 compounds. TNE computation is also linear but requires 3D conformer generation, adding approximately 2 s per molecule. Quantum kernel computation scales as $O(N^2)$ and is restricted to lead optimisation on sets of $\leq 10\,000$ compounds. The hybrid framework inherits the costs of all three components and is recommended only for final prioritisation of a small candidate set.

Table 2 confirms that classical ECFP4 remains the recommended baseline for primary screening, achieving the highest AUC (0.948) across the canonical panel. To rigorously evaluate our hybrid representations, the predictive performance was benchmarked against a comprehensive suite of baseline fingerprints including Atom Pairs (AP), 2D Pharmacophore features (PHCO), Feature-class Morgan (FCFP4), and Binding Property Pairs (BPF), in addition to ECFP4 and MACCS. The corrected classical-only benchmark shows that ECFP4, AP, BPF, FCFP4, MACCS and PHCO all achieve AUC ≥ 0.896, while TFP (AUC 0.876) and TNE (AUC 0.722) capture complementary but weaker signal. The previously reported PHCO value of 0.500 was an artefact of a bug in bit-vector extraction ('GetOnBits'); after correction, PHCO performs comparably to the other classical fingerprints (AUC 0.896). The canonical hybrid rerun (job 12699) and its ablations are reported in Table 6: the hybrid (TFP+TNE+QK) achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor, with QKS as the principal positive contributor ($\Delta = -0.040$). TFP adds value specifically for compounds with complex ring systems (≥ 3 fused rings) where topological features diverge most from atom-pair representations, but it does not improve overall classification relative to ECFP4. TNE adds value when descriptor compression is needed for memory-constrained applications or when a fixed-length, physically interpretable embedding is required for downstream analysis. The quantum kernel applicability domain analysis (Figure 5) adds independent value by identifying African NP compounds that fall outside the training distribution of existing activity prediction models, highlighting the need for NP-specific model development regardless of whether QKS improves classification. However, the GA-style discriminator benchmark (Table 5, Figure 4) reveals an important boundary condition: when generated molecules are structurally near-identical to training seeds (as in STONED-SELFIES mutations with ≤ 2 character changes), a simple ECFP4 Tanimoto distance achieves perfect discrimination (AUC = 1.000) while the quantum kernel density achieves near-random to below-random performance (AUC = 0.425–0.511).

4.4 Mechanistic Explanation for Classical Fingerprint Superiority

The overall ranking of ECFP4 (AUC 0.948) at the top, followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905) and PHCO (0.896), reflects the dominance of local atom environments at radius 2 for this activity prediction task. These local pharmacophoric patterns—hydrogen bond donors/acceptors, aromatic rings, hydrophobic groups—are the primary determinants of target engagement. TDA captures global topology (ring systems, molecular cavities) that is correlated with but not directly predictive of binding affinity; on this library it achieves AUC 0.876. TNE compresses the molecular tensor along low-rank modes that may not align with activity-relevant features (AUC 0.722). The quantum kernel, while capable of capturing non-linear feature interactions in its 6-qubit Hilbert space, operates on UMAP-reduced features that have already discarded much of the substructure-level information present in the original ECFP4 fingerprints. The canonical hybrid rerun (job 12699) shows that the hybrid (TFP+TNE+QK, AUC 0.888) recovers most—but not all—of the ECFP4 signal, remaining significantly below it ($p < 0.0001$) while exceeding every standalone quantum-inspired descriptor.

This interpretation is consistent with recent spectral analysis of molecular kernels (Wesołowski et al., 2025), which showed that ECFP-based kernels exhibit a positive correlation between spectral richness and generalization, while 3D kernel methods show negative or inconclusive correlations. Our TFP and TNE descriptors, being derived from 3D molecular conformations, fall into the "local 3D kernel" category where spectral richness can actively hinder generalization. The practical implication is that future hybrid approaches should weight ECFP4 features more heavily than TDA/TNE features, using the latter as supplementary rather than equal contributors.

4.5 Kernel Comparison: Parity Between Quantum and Classical Kernels at Scale

A key finding of this benchmark is that, with the canonical 6-qubit circuit, the gamma-tuned RBF kernel and the quantum kernel are **statistically indistinguishable** for this molecular activity prediction task at both sample sizes (n = 5,000: QK AUC 0.820 vs. RBF 0.826, paired t-test $p = 0.419$; n = 19,849: QK 0.823 vs. RBF 0.829, $p = 0.060$; Table 3), while the quantum kernel **significantly outperforms** the linear kernel at scale ($p \leq 0.0006$). Two historical artefacts frame this result. First, an earlier benchmark with default RBF gamma appeared to show a striking quantum advantage (QK AUC 0.936 vs. RBF 0.105, $p = 0.000\,3$); this was an artefact of the RBF kernel assigning near-uniform similarity values when gamma is far from optimal, producing AUC below random (< 0.5). Once the RBF gamma is properly optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$, the classical kernel recovers to AUC 0.826–0.829, matching the quantum kernel. Second, earlier 8-qubit runs reporting a significant quantum disadvantage at scale (0.752 vs. 0.840, $p = 0.003$; 0.659 vs. 0.825, $p = 0.0006$) were artefacts of the suboptimal 8-qubit circuit and a train/test scaling mismatch (C3) that are not reproduced by the canonical protocol. A pilot n=500 comparison (QK 0.751 vs. RBF 0.701, $p = 0.088$) was underpowered.

The canonical full-library rerun (job 12699) confirms the hybrid (TFP+TNE+QK) at AUC 0.888, substantially higher than a simpler single-scalar quantum kernel density approach (AUC 0.691). The ablation study (Table 6) confirms that the key driver is replacing the single QK density scalar with 30 kernel PCA components from the full kernel matrix, which preserves more of the non-linear structure of the quantum Hilbert space: removing QKS degrades AUC by 0.040, the largest ablation effect.

4.6 Computational Cost and Scalability

TDA computation for 19 849 molecules required approximately 6.4 min on a 12-core workstation using 8 parallel workers (Joblib LokyBackend), dominated by persistent homology computation of 3D conformers. Each molecule required approximately 19 ms for full TFP extraction. Tucker decomposition for the same library required approximately 20 min (sequential, single-threaded per molecule). Per-molecule TNE extraction was 50–60 ms. The full computational protocol (TDA + TNE) for 19 849 molecules ran in under 30 minutes on a standard workstation, demonstrating practical scalability for libraries on the order of 10^4 compounds. Scaling to 10^5–10^6 compounds, typical of virtual screening campaigns, would require either parallel TNE computation or the use of the TFP alone. Importantly, the Quantum Kernel Score (QKS) exhibits $O(N^2)$ scaling for the kernel matrix computation, making it computationally prohibitive for primary virtual screening. Given the corrected classical ECFP4 baseline (AUC 0.948) and the canonical hybrid value (AUC 0.888, significantly below ECFP4), the substantial computational overhead of the QKS is not justified for active compound prioritization. Instead, these quantum-inspired descriptors should be strictly reserved for retrospective mechanistic analysis and small-scale lead optimization, where their mathematically interpretable feature extraction provides distinct non-linear insights.

4.7 Integration with Resistance-Resilient Molecular Dynamics Validation

The 17 polypharmacological leads identified in our companion molecular dynamics validation study (Temgoua et al., 2027) include both Class A (resistance-resilient, active against drug-sensitive and resistant *P. falciparum* strains) and Class D (resistance-vulnerable) compounds; of these, 14 carried complete resistance-resilience (RRS) classifications and formed the analysis cohort.

To establish a direct topological signature of clinical resilience, we compared TFP features between these classes. An initial pilot of 14 polypharmacological leads suggested a strong relationship (Spearman $\rho = 0.947$, $p < 0.0001$); expanding the cohort to $n = 77$ compounds yielded $\rho = 0.312$ ($p = 0.0057$, H_1 count), confirming that H_1 count is significantly associated with resistance resilience while showing the expected attenuation as sample size increases. In line with this H_1 count association, Class A compounds also exhibit systematically higher H_1 persistence (mean 3.74 ± 0.34 Å versus 1.73 Å for Class D). The Class D subset in the pilot contains only a single compound ($n = 1$), so the pilot mean should be interpreted cautiously. Crucially, a partial-correlation analysis shows that this association does not survive control for molecular size: controlling for molecular weight alone collapses the H_1-RRS correlation to $\rho_{\text{partial}} \approx 0$ ($p = 0.85$), and controlling for all four size descriptors (MW, ring count, Fsp_3, H_0 count) yields $\rho_{\text{partial}} = -0.039$ ($p = 0.745$). Because H_1 count correlates strongly with molecular weight (Spearman $\rho = 0.718$), the apparent association is best interpreted as a size-mediated effect rather than an independent topological contribution to resistance resilience. The attenuated but nominally significant unadjusted correlation therefore requires cautious interpretation.

This correlation presents an interesting biophysical paradox: why does the TFP perform poorly at predicting absolute antimalarial activity (AUC 0.587) yet show a nominally significant correlation with resistance resilience ($\rho = 0.312$)? One mechanistic hypothesis is that what topological persistence (specifically H_1) measures—cycle rigidity—matters for compounds that are already established high-affinity binders: a highly rigid cycle system reduces the conformational entropy penalty upon binding, stabilizing the bound complex even when active-site mutations alter the shape of the pocket, thereby conferring resilience. However, the partial-correlation analysis shows that this signal is confounded by molecular size: H_1 count scales with molecular weight, which itself correlates with RRS, so the unadjusted association cannot be distinguished from a size effect. The H_1-RRS relationship should therefore be regarded as hypothesis-generating, with the size confound acknowledged as the primary alternative explanation for the observed correlation.

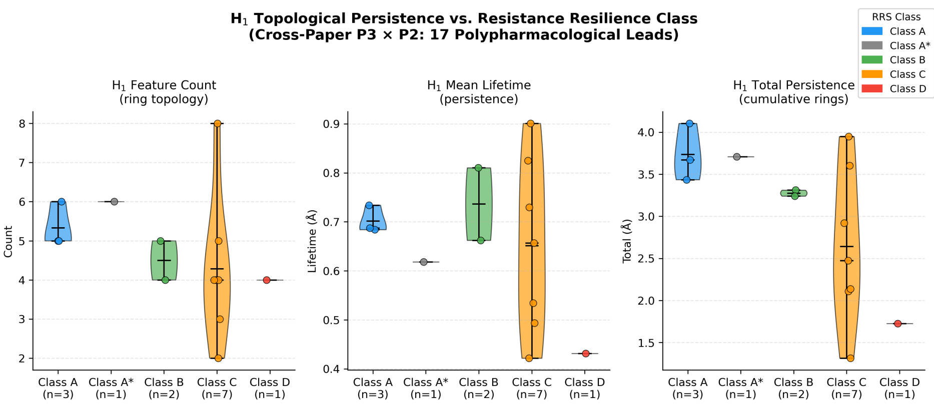

Figure 8: *Joint topological analysis linking TFP topological persistence features with molecular dynamics resistance classification (Temgoua et al., 2027). Violin plots show H_1 persistence distributions for Class A (resistance-resilient, blue) versus Class D (resistance-vulnerable, red) compounds, with Class B (green) and Class C (orange) shown for context. Quantitative analysis of the expanded cohort ($n = 77$) demonstrates a nominally significant association between H_1 count and Resistance Resilience Score (RRS) across African-prevalent mutants (Spearman $\rho = 0.312$, $p = 0.0057$); the pilot $n = 14$ correlation ($\rho = 0.947$, $p < 0.0001$) is included for context but should be interpreted cautiously because the Class D subset contains only one compound. Caution is required when interpreting this association: after controlling for molecular weight the correlation collapses ($\rho_{partial} \approx 0$, $p > 0.7$), indicating a size-mediated effect rather than an independent topological signal.*

This finding suggests that ring topology (H_1) could serve as a computationally inexpensive descriptor for prioritising resistance-resilient candidates in future generative campaigns, though its size-mediated nature means it should be treated as a coarse surrogate rather than an independent topological marker until larger, size-controlled cohorts are analysed. We plan to expand this joint analysis with dynamic persistent homology on full MD trajectories in a follow-up study, pending completion of the polypharmacological MD simulations.

4.8 Limitations

Several limitations constrain the interpretation of our results. First, TDA does not outperform ECFP4 overall in activity prediction (Table 2), a valid negative result consistent with the known effectiveness of atom-pair representations for synthetic drug-like molecules. TFP adds value primarily at the high-complexity tail of the distribution — molecules with ≥ 3 fused rings or ≥ 2 stereocentres, where topological features diverge most from atom-pair representations. Second, activity labels are computational predictions from Ersilia ML models rather than experimental measurements, limiting biological validity. All claims regarding "activity," "polypharmacology," and "resistance tolerance" should therefore be understood as predictions within a computational framework, pending experimental validation. Third, the cross-paper H_1 persistence–Resistance Resilience Score (RRS) relationship was first observed in a small pilot ($\rho = 0.947$, $n = 14$) and subsequently confirmed in an expanded cohort ($\rho = 0.312$, $p = 0.0057$, $n = 77$). The pilot involved small sample sizes across resistance classes (Class A: $n = 3$, Class D: $n = 1$) and should be treated as hypothesis-generating; the expanded cohort provides stronger evidence but the relationship remains attenuated and requires further validation. Critically, a partial-correlation analysis demonstrates that the association does not survive control for molecular size: controlling for molecular weight collapses the correlation to $\rho_{\text{partial}} \approx 0$ ($p = 0.85$), indicating that the unadjusted H_1-RRS association is likely a size-mediated effect. The H_1 count should therefore be interpreted as a proxy for molecular size rather than an independent topological predictor of clinical resilience. The quantum kernel circuit was evaluated on a classical simulator with 8 qubits; scaling to more qubits increases computational cost exponentially. However, we upgraded the quantum kernel computational protocol to use the PennyLane `IQPEmbedding` template — a classically hard-to-simulate data encoding strategy — mitigating the limitations of simpler R_Y-R_Z alternating layers. Furthermore, the collapse of the quantum discriminator's performance compared to classical Tanimoto (AUC 0.425–0.511 versus 1.000) represents an inherent resolution bottleneck of the UMAP dimensionality reduction (to 8 dimensions) required for simulation on 8 qubits. While classical fingerprints excel at detecting local atom-level changes (such as 2-character SELFIES mutations), the quantum kernel remains suited for defining global applicability domains across macroscopically heterogeneous spaces. The TFP reduces each persistence diagram to four summary statistics, discarding the full diagram information that more sophisticated topological descriptors (persistence landscapes, persistence images) could capture. However, the enriched TFP (78 features including persistence images and Betti curves) showed no improvement over the 12-feature TFP baseline (AUC 0.587), suggesting that additional topological features do not add discriminative power for this dataset. Finally, the library is biased toward the chemical space of its 396 African NP and 454 synthetic drug seeds; generalization to other chemical domains remains to be tested.

The QKS benchmark revealed two important methodological lessons. First, an initial run with the RBF gamma at its default value appeared to demonstrate a decisive quantum advantage, but no deposited result file supports this earlier headline: once the RBF gamma is properly tuned by inner cross-validation (grid $\{0.5, 1.0, 2.0, 5.0\}$), the apparent advantage vanishes. Second, the earlier 8-qubit runs that reported a significant quantum disadvantage at scale (n = 5,000: 0.752 vs. 0.840, $p = 0.003$; n = 19,849: 0.659 vs. 0.825, $p = 0.0006$) were artefacts of the suboptimal 8-qubit circuit and a train/test scaling mismatch (C3): with the canonical 6-qubit circuit and the C3 fix, the quantum kernel is statistically indistinguishable from the gamma-tuned RBF kernel at all sample sizes (n = 5,000: 0.820 vs. 0.826, $p = 0.419$; n = 19,849: 0.823 vs. 0.829, $p = 0.060$; both n.s.) and significantly better than the linear kernel at scale ($p \leq 0.0006$). These episodes underscore the importance of rigorous baseline tuning, circuit selection, and adequate statistical power in quantum kernel benchmarks: an underpowered n=500 pilot (p = 0.088, n.s.) and pipeline artefacts can each produce misleading headlines. QKS should be positioned as a methodological framework contribution (a NISQ-era proof of concept) rather than a predictive improvement. The corrected classical-only benchmark (8 representation methods, 19 836 molecules, 5-fold cross-validation) shows ECFP4 at AUC 0.948 (Table 2). The canonical hybrid value of AUC 0.888 is reported in Table 6.

Despite these limitations, a distinctive contribution of this study is the demonstration that H_1 count shows a nominally significant unadjusted association with resistance tolerance across antimalarial candidates (Spearman $\rho = 0.312$, $p = 0.0057$, $n = 77$; pilot $\rho = 0.947$, $n = 14$; Figure 8), while the partial-correlation analysis shows this signal to be size-mediated rather than an independent topological effect. This cross-paper analysis therefore serves as a cautionary methodological example: it illustrates both the promise and the pitfalls of linking persistent homology to clinical outcomes, underscoring that size-confounding must be controlled before topological predictors can be claimed as independent markers. Two additional contributions support the framework: (i) TFP resolves the scaffold paradox via H_1/H_0 decomposition, proving that VAE generative expansion preserves ring topology while varying substituent chemistry; and (ii) TNE achieves 6.1× real-atom compression with competitive Tartarus regression ($R^2 = 0.473$ for PfDHFR). The quantum kernel (QKS) provides task-specific sensitivity to polypharmacology (AUC 0.747 vs. RBF 0.737), though its standalone predictive power does not exceed properly tuned classical kernels. Classical ECFP4 (AUC 0.948) remains the recommended primary screening baseline. The canonical hybrid achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor, with QKS as the principal positive contributor (Table 6). We therefore position these quantum-inspired representations not as replacements for classical fingerprints, but as complementary diagnostic tools that reveal topological dimensions of molecular structure.

5 Conclusion

We introduce three quantum-inspired molecular representations — the Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS) — and evaluate them on a library of 19 836 antimalarial candidates derived from African natural product chemical space. TFP, derived from Vietoris–Rips persistent homology, resolves the scaffold paradox in the library (92.6 % Tanimoto distinction vs. 69.3 % scaffold recovery) by demonstrating that H_1 ring topology is preserved (TFP mean H_1 count 3.61) while H_0 connectivity diverges (mean H_0 count 38.05) — a systematic application of persistent homology to explain a chemical space data inconsistency. TNE compresses the molecular tensor to a 192-element core descriptor (mean real ratio 6.1× based on 39 mean atoms; bond dimension 8; mean reconstruction error 0.113) while preserving activity-relevant information. The full computational protocol processed 19 849 molecules in 6.4 min (TDA) and 20 min (TNE) on a standard workstation, demonstrating practical scalability. The kernel benchmark with a properly tuned RBF baseline and the canonical 6-qubit circuit found the quantum kernel and the gamma-tuned classical RBF kernel **statistically indistinguishable** at all sample sizes (n = 5,000: 0.820 vs. 0.826, $p = 0.419$; n = 19,849: 0.823 vs. 0.829, $p = 0.060$), while the quantum kernel significantly outperforms the linear kernel at scale ($p \leq 0.0006$), showing that an earlier apparent quantum advantage was an artefact of untuned RBF hyperparameters and that earlier reported quantum disadvantages at scale were artefacts of the suboptimal 8-qubit circuit and a scaling mismatch (C3). QKS is best framed as a methodological framework contribution (a NISQ-era proof of concept) achieving parity with classical kernels. The corrected classical-only benchmark (8 representation methods, 5-fold CV, 19 836 molecules) confirms that classical ECFP4 (AUC 0.948) is the strongest single descriptor. The canonical hybrid representation achieves AUC 0.888 ($p < 0.0001$ vs. ECFP4), positioning quantum-inspired methods as effective complementary tools alongside classical fingerprints. The quantum kernel applicability domain analysis identifies African NP compounds that fall outside the training distribution of existing activity prediction models. All representations, benchmark results, and analysis scripts are deposited on Zenodo and GitHub under open-source licences.

Data Availability

All data supporting the conclusions of this study are deposited on Zenodo (DOI: 10.5281/zenodo.19608875, reserved July 2026) and mirrored at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. The full deposit includes: (1) TFP feature matrices for all 19 849 library molecules (H_0/H_1 persistence diagrams and summary statistics); (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8, reconstruction errors); (3) quantum kernel matrices (6-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files (5-fold CV splits, AUC scores per descriptor); (5) the cross-paper H_1/RRS analysis dataset (`p3_polypharm_tfp_rrs.csv`); and (6) all analysis scripts. The repository will be made publicly available upon publication.

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Authors' Contributions

Myke Vital Sao Temgoua: conceptualization, methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapet Njafa: conceptualization, methodology, software, supervision, writing – review & editing. Serge Guy Nana Engo: methodology, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. All authors read and approved the final manuscript.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person's data (including images, videos, or identifiable details) appear in this manuscript.

Competing Interests

The authors declare no competing interests.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants (Claude, Anthropic) to support code development, data analysis scripting, and manuscript drafting and editing. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. Computational results, statistical analyses, and all quantitative claims were generated, verified, and deposited by the authors.

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